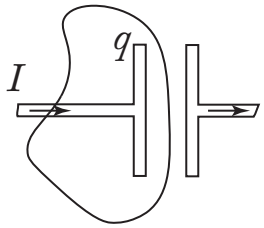
ELECTROMAGNETISM: ANSWERS

This is a closed-book examination. A formula sheet is provided

Question 1. [a) = 10], [b) = 10], [c) = 5], [d) = 10], [e) = 5]

a) (bookwork)



Choose a volume V that encloses one plate of the capacitor and the short section of wire connected to it, and let its surface S cut the wire once [2]. Then the surface integral gives the conduction current through the wire:

$$-\oiint_S \mathbf{J}\cdot d\mathbf{s} = I. \qquad [3]$$

The negative sign shows that the current flows into S. Using the integral form of the continuity equation, we obtain

$$I = \frac{dq}{dt}. \qquad [1]$$

Let U be the capacitor voltage (potential difference between the plates). By definition of capacitance, $C = q/U$, hence $q = CU$. [2]

Therefore

$$I = \frac{d}{dt}(CU) = C\frac{dU}{dt}. \qquad [2]$$

b) (bookwork)

Using the line-integral definition of potential difference,

$$\varphi(B) - \varphi(A) = -\int_A^B \mathbf{E}\cdot d\mathbf{l}. \qquad [3]$$

Take any two points A and B on the surface of the PEC, and choose a path that lies entirely on the surface. Along this path, $d\mathbf{l}$ is tangential to the surface. [2]

On a PEC surface, $\mathbf{E}_\tau = \mathbf{0}$, so $\mathbf{E}\cdot d\mathbf{l} = 0$ everywhere along the surface path. [3] Hence the integral is zero and

$$\varphi(B) - \varphi(A) = 0, \qquad [2]$$

Therefore the PEC surface is an equipotential surface.

c) (calculated example)

Using an expression from the formula sheet for the input impedance of a loaded lossless line,

$$Z_{\text{in}}(d) = Z_0 \frac{Z_L + jZ_0 \tan(kd)}{Z_0 + jZ_L \tan(kd)}. \qquad [2]$$

Since $k = 2\pi/\lambda$, we have $k(d + \lambda/2) = kd + \pi$, and $\tan(k(d + \lambda/2)) = \tan(kd + \pi) = \tan(kd)$. [1] Substituting into the expression above gives

$$Z_{\text{in}}(d + \lambda/2) = Z_{\text{in}}(d). \qquad [1]$$

Interpretation: shifting the observation point by $\lambda/2$ repeats the standing-wave pattern; each of the forward and reflected waves acquires an additional phase factor $e^{-jk(\lambda/2)} = e^{-j\pi} = -1$, so their ratio (and thus V/I) is unchanged [1].

d) (calculated example)

For a one-port referenced to Z_0, write the total voltage and current at the reference plane in terms of forward and reflected waves:

$$S_{11} = \frac{V_1 - Z_0 I_1}{V_1 + Z_0 I_1} = \frac{V_1/I_1 - Z_0}{V_1/I_1 + Z_0}. \qquad [2]$$

However, $V_1/I_1 = Z$ [2], leading to

$$S_{11} = \frac{Z - Z_0}{Z + Z_0}$$

Rearrange to get

$$Z = Z_0 \frac{1 + S_{11}}{1 - S_{11}}. \qquad [3]$$

If Z is a reactance, $Z = jX$, then

$$|S_{11}| = \left| \frac{jX - Z_0}{jX + Z_0} \right| = \frac{\sqrt{X^2 + Z_0^2}}{\sqrt{X^2 + Z_0^2}} = 1$$

A purely reactive impedance stores but does not dissipate time-averaged power, so all incident power is reflected [3].

e) (bookwork)

A short dipole ($d \ll \lambda$) has an approximately uniform current along its length [2]. A half-wave dipole has a standing-wave (approximately sinusoidal) current distribution: maximum at the centre and zero at the ends [2]. This difference arises because the half-wave dipole is resonant, whereas the short dipole is electrically small and cannot support a sinusoidal standing wave along its length [1].

Question 2. [a) = 5], [b) = 10], [c) = 5], [d) = 10]

a) (bookwork)

Assume harmonic time dependence and write $\mathbf{E}(\mathbf{r},t) = \mathrm{Re}\{\mathbf{E}(\mathbf{r})e^{j\omega t}\}$. Then

$$\frac{\partial^2 \mathbf{E}}{\partial t^2} = -\omega^2 \mathbf{E}. \qquad [2]$$

Assume propagation in the z-direction only, so $\mathbf{E}$ depends on z only and $\nabla^2 \mathbf{E} = d^2\mathbf{E}/dz^2$ [2]. Substituting into $\nabla^2 \mathbf{E} - \mu_0 \varepsilon_0 \, \partial^2 \mathbf{E}/\partial t^2 = 0$ gives

$$\frac{d^2 \mathbf{E}}{dz^2} + \omega^2 \mu_0 \varepsilon_0 \, \mathbf{E} = 0. \qquad [1]$$

b) (bookwork)

The equation in part (a) is a homogeneous second-order ODE. The general solution is of the form

$$\mathbf{E}(z) = \mathbf{E}^+ e^{-jkz} + \mathbf{E}^- e^{+jkz}. \qquad [3]$$

The two terms correspond to two counter-propagating waves [1]. The dispersion relation (characteristic polynomial) is $\omega = ck$ [3], where $c = 1/\sqrt{\varepsilon_0 \mu_0}$ [3].

c) (bookwork)

A transverse electromagnetic (TEM) wave has both $\mathbf{E}$ and $\mathbf{H}$ perpendicular (transverse) to the direction of propagation, and a linearly polarised wave has $\mathbf{E}$ oscillating in a fixed transverse direction [3].

For propagation in $+z$ with linear polarisation along x,

$$\mathbf{E}(z,t) = E_0 \, \mathbf{e}_x \, e^{j(\omega t - kz)} \qquad [2]$$

d) (bookwork/calculated example)

Using Maxwell's equation for harmonically varying fields,

$$\nabla \times \mathbf{E} = -j\omega \mathbf{B} = -j\omega \mu_0 \mathbf{H}. \qquad [2]$$

For $\mathbf{E} = E_0 \, \mathbf{e}_x \, e^{-jkz}$ we have $E_x = E_0 e^{-jkz}$ and

$$\nabla \times \mathbf{E} = \left(\frac{\partial E_x}{\partial z} \right) \mathbf{e}_y = (-jkE_x) \, \mathbf{e}_y. \qquad [3]$$

Therefore

$$-jkE_x \, \mathbf{e}_y = -j\omega \mu_0 \mathbf{H} \;\Rightarrow\; \mathbf{H} = \frac{k}{\omega \mu_0} E_x \, \mathbf{e}_y. \qquad [3]$$

Using $k = \omega \sqrt{\mu_0 \varepsilon_0}$ gives

$$\frac{k}{\omega \mu_0} = \sqrt{\frac{\varepsilon_0}{\mu_0}} = \frac{1}{Z_0}, \qquad [1]$$

hence

$$\mathbf{H}(z,t) = \frac{E_0}{Z_0} \, \mathbf{e}_y \, e^{j(\omega t - kz)} \qquad [1]$$

Question 3. [a) = 15], [b) = 10], [c) = 5]

a) (bookwork/calculated example)

Using the time-averaged Poynting vector for harmonic fields,

$$\mathbf{P} = \frac{1}{2}\mathrm{Re}\{\mathbf{E} \times \mathbf{H}^*\}. \qquad [3]$$

In the far field of the short dipole, $E_r = 0$ and the only non-zero components are E_θ and H_ϕ, so

$$\mathbf{E} \times \mathbf{H}^* = (E_\theta\, \mathbf{e}_\theta) \times (H_\phi^*\, \mathbf{e}_\phi) = E_\theta H_\phi^*\, \mathbf{e}_r, \qquad [3]$$

i.e. the power flow is radial (outward for radiation). Using $H_\phi = E_\theta / Z_0$,

$$\mathbf{P} = \frac{1}{2}\mathrm{Re}\left\{E_\theta \left(\frac{E_\theta^*}{Z_0}\right)\right\}\mathbf{e}_r = \frac{|E_\theta|^2}{2Z_0}\,\mathbf{e}_r. \qquad [3]$$

so

$$\mathbf{P} = \frac{Z_0 |I_0|^2 k^2 d^2}{32\pi^2}\,\frac{\sin^2\theta}{r^2}\,\mathbf{e}_r. \qquad [3]$$

The $1/r^2$ dependence follows because $E_\theta, H_\phi \propto 1/r$ in the far field, so $\mathbf{P} \propto 1/r^2$, giving constant total radiated power through a sphere of radius r (since area $\propto r^2$) [3].

b) (bookwork)

P_r is the received power delivered to the receive load. [1]

P_t is the power delivered to the transmit antenna. [1]

G_t is the transmit antenna gain in the direction of the receive antenna. [1]

G_r is the receive antenna gain in the direction of the transmit antenna. [1]

λ is the free-space wavelength at the operating frequency. [1]

R is the separation distance between antennas. [1]

Physical meaning: in free space (far field), the receive power is proportional to the transmit power and the transmitter and receiver antenna gains [2]. The factor $(\lambda/4\pi R)^2$ expresses spherical spreading of power density ($\propto 1/R^2$) together with the frequency dependence of the effective area ($\propto \lambda^2$) [2].

c) (calculated example)

From Friis' equation, with G_t, G_r fixed,

$$P_r \propto P_t \left(\frac{\lambda}{R}\right)^2. \qquad [1]$$

Doubling frequency halves wavelength: $\lambda \mapsto \lambda/2$ [1]. Doubling separation gives $R \mapsto 2R$. Therefore the propagation factor changes by

$$\left(\frac{\lambda/2}{2R}\right)^2 \Big/ \left(\frac{\lambda}{R}\right)^2 = \left(\frac{1}{4}\right)^2 = \frac{1}{16}.$$

To keep P_r unchanged, P_t must be increased by a factor of 16 [3].